Cache Performance

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Hit: When the CPU refers to the memory and reveals the word cache, it's far stated that a **hit** has successfully occurred.

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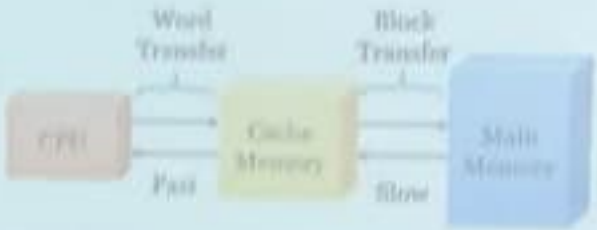
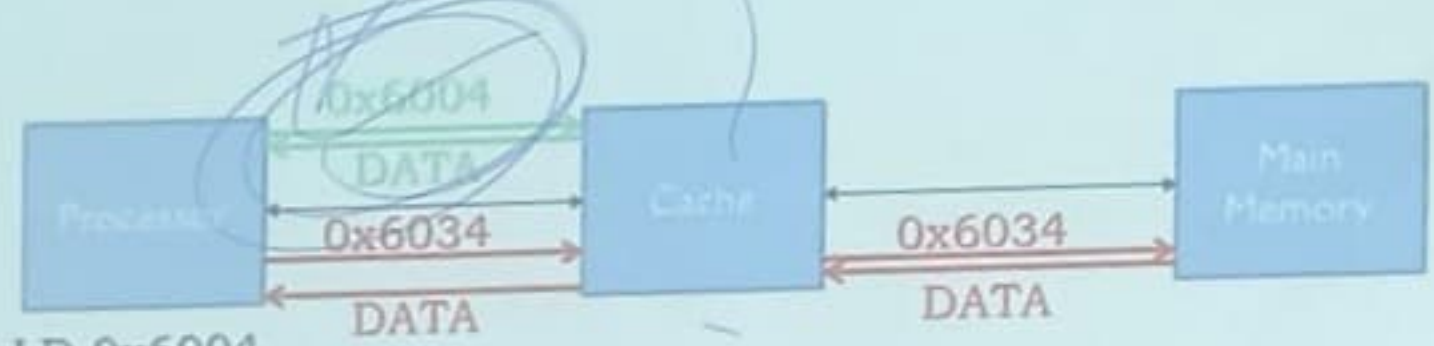


Cache Performance



LD 0x6004
LD 0x6034

CACHE HIT

CACHE MISS

Hit Ratio

- The **hit ratio H,** is nothing but a probability of getting hits out of some number of memory references made by the CPU.
- So its range is $0 \leq H \leq 1$.

Hit Ratio = *Number of Hits / Total CPU references to memory*

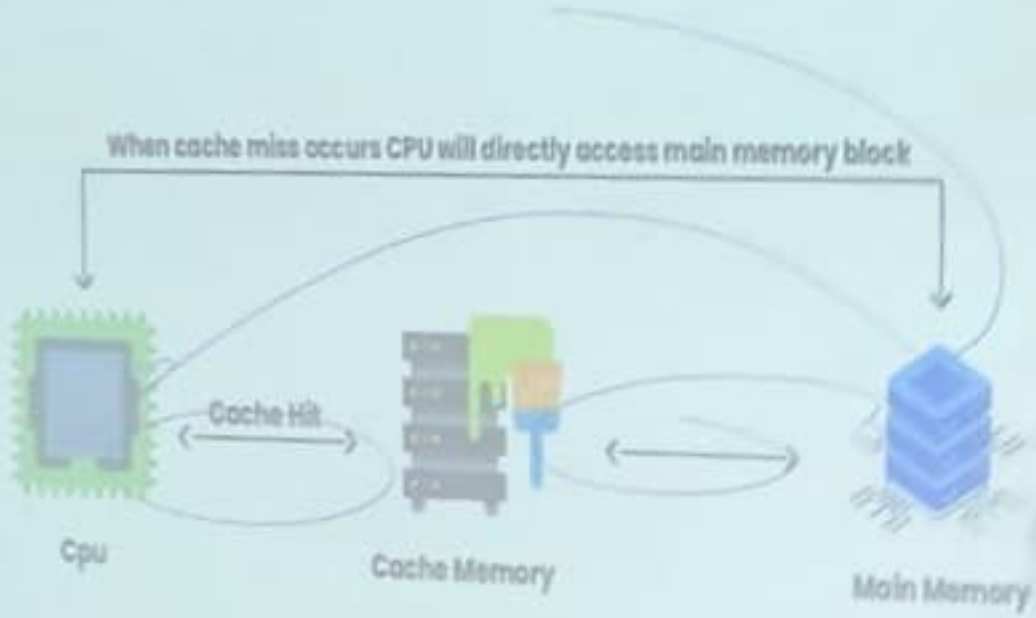
= *Number of hits / (Number of Hits + Number of Misses)*

Hit Ratio

$$\frac{\text{Cache Hits}}{\text{Total Cache Request}} = \text{CACHE HIT RATIO}$$

(Cache Hits + Cache Misses)

AMAT or EMAT


When cache miss occurs CPU will directly access main memory block
Cache Hit
Cpu
Cache Memory
Main Memory

AMAT

- In order to look at the performance of cache memories, we need to look at the *average memory access time* (**AMAT**) and the factors that will affect it.

$$\text{AMAT} = \text{Hit Time} + \text{Miss Time}$$
$$= \text{Hit Time} + \text{Miss Rate} \times \text{Miss Penalty}$$

- **Miss Rate :** the fraction of accesses that are not in the cache
  - (1-H).
- **Miss Penalty :** the additional clock cycles to service the miss
  - the extra time needed to carry the favored information into cache from main memory in case of miss in cache.

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Cache Performance

- The performance of a single-level cache system for a read operation can be characterized by the following equation:

$$T_a = HT_c + (1 - H) \times (T_m + T_c)$$
$$= T_c + (1 - H) \times T_m$$

- Here

  T_a = the average memory access time (AMAT),

  T_c = the cache access time,

  T_m = the memory access time (e.g., memory to processor register),

  H = the hit ratio.

EMAT

Cache Performance

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- Here

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T_c = the cache access time,
T_m = the memory access time (e.g., memory to processor register),
H = the hit ratio.

Hit Time Reduction Techniques

- Small and simple cache
- Avoid address translation in indexing cache, i.e., address translation
- Pipelined cache access
- Trace caches

Miss Rate Reduction Techniques

- Larger block size
- Increase cache size
- Higher associativity
- Way prediction
- Pseudo-associative caches
- Compiler optimizations

Miss Penalty Reduction Techniques

- Multilevel cache
- Critical word first and early restart
- Priority to read miss
- Merging write buffer
- Victim cache

Miss Penalty X Rate Reduction Techniques

- Non-blocking cache
- Hardware prefetching
- Compiler-controlled prefetching

Advanced Optimization Techniques

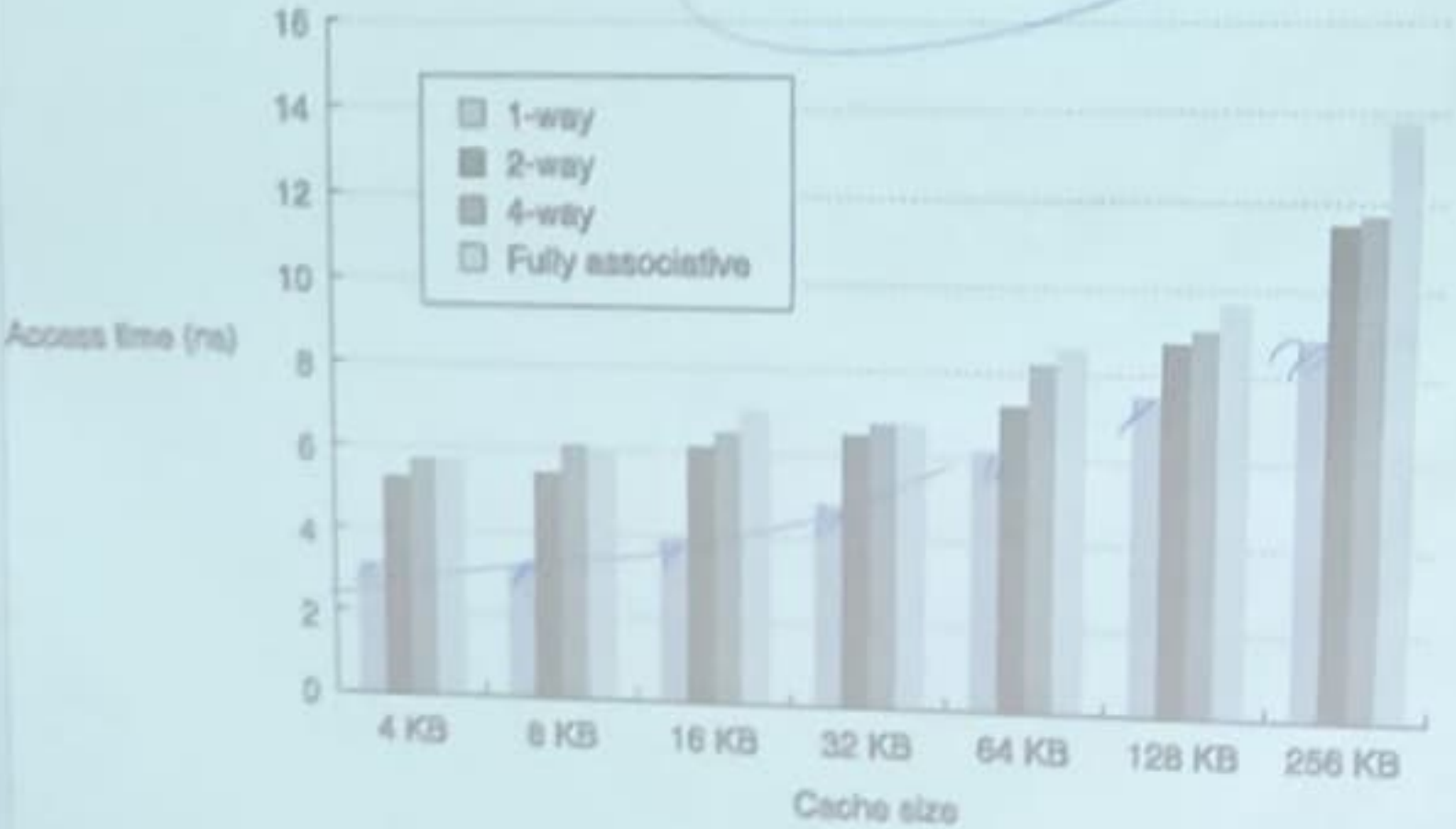
- Pipelined caches
- Multi-banked caches
- Non-blocking caches

Access Time

Access Time

Access time estimate for 90 nm using CACTI model 4.0

Access Time in a CMOS Cache

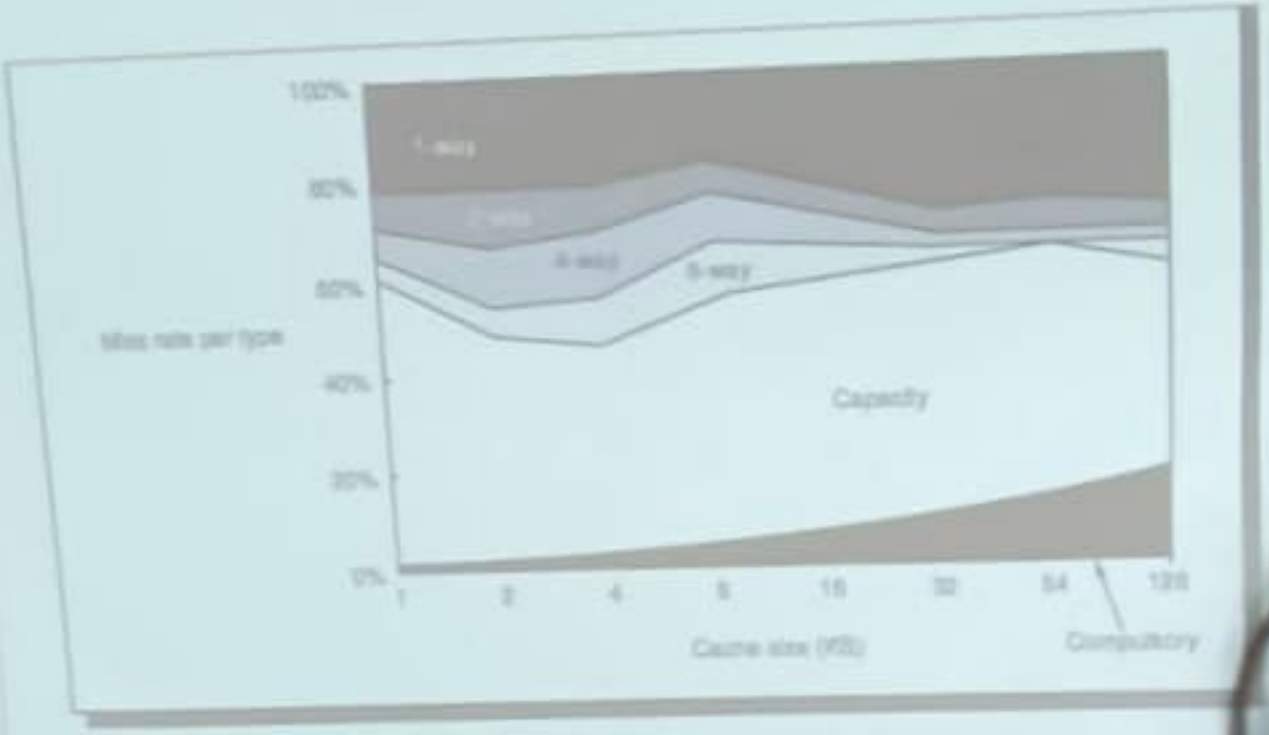


Reducing Miss Rate

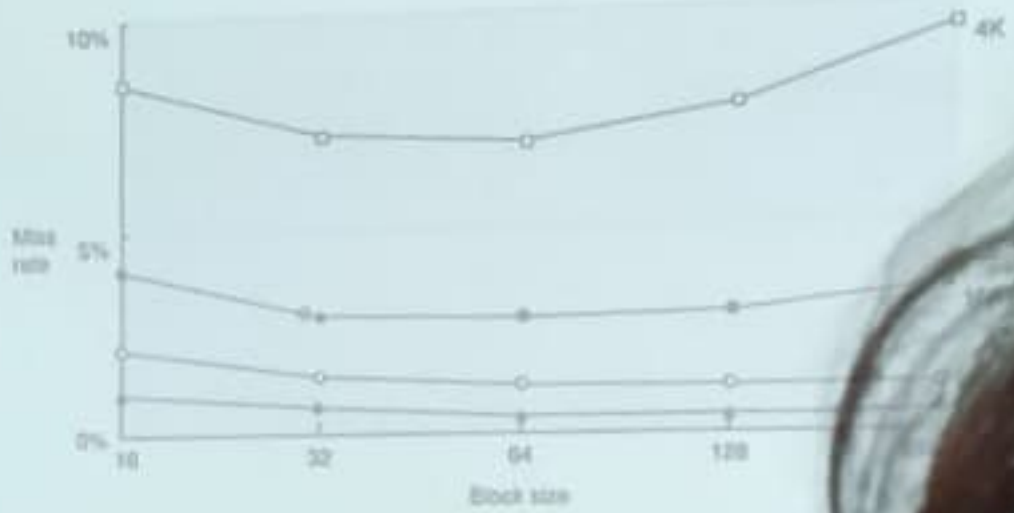
Three Types of Misses

- *Compulsory*
  - During a program, the very first access to a block will not be in the cache (unless pre-fetched)
- *Capacity*
  - The *working set* of blocks accessed by the program is too large to fit in the cache
- *Conflict*
  - Unless cache is fully associative, sometimes blocks may be evicted too early because too many frequently-accessed blocks map to the same limited set of frames.

As Fraction of Total Misses

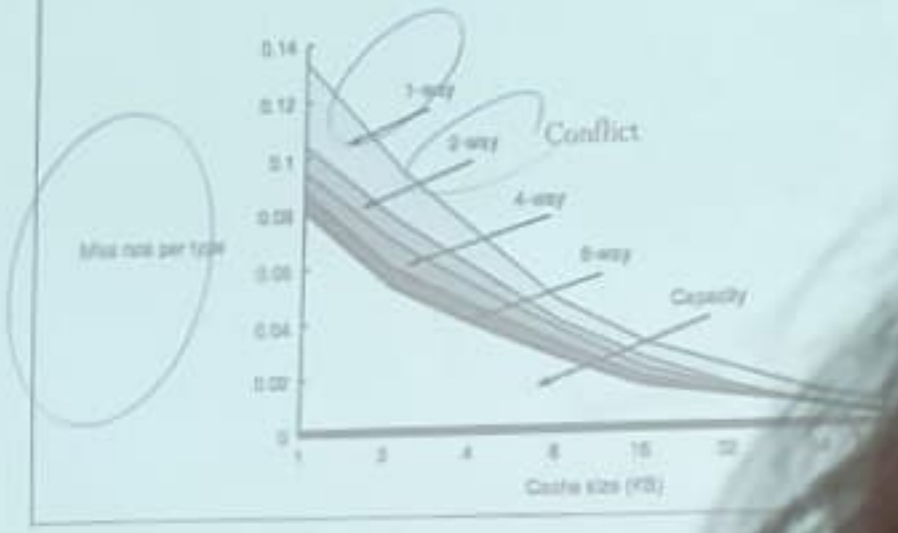


Block Size Effect



Miss rate is actually goes up if the block is too large relative to the cache

Misses by Type



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- From direct-mapped to 2-way helps as much as doubling
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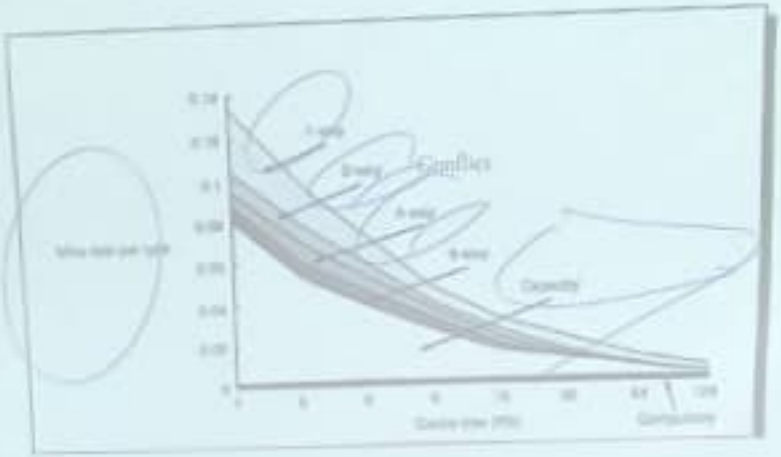

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Performance Comparison

- Assume:

$$\text{Hit Time}_{\text{one}} = \text{Cycle Time}_{\text{one}} \qquad \text{Miss Penalty} = 25 \times \text{Cycle Time}_{\text{one}}$$
$$\text{Hit Time}_{\text{two}} = 1.36 \times \text{Cycle Time}_{\text{one}}$$
$$\text{Hit Time}_{\text{four}} = 1.44 \times \text{Cycle Time}_{\text{one}}$$
$$\text{Hit Time}_{\text{eight}} = 1.52 \times \text{Cycle Time}_{\text{one}}$$

- 8KB, 1-way miss rate=9.8%, 4-way miss rate=7.1%

$$\text{Average Mem Access Time}_{\text{one}} = 1.00 + \text{Miss Rate} \times 25$$
$$= 1.00 + 0.098 \times 25 = 3.45$$

$$\text{Average Mem Access Time}_{\text{four}} = 1.44 + \text{Miss Rate} \times 25$$
$$= 1.44 + 0.071 \times 25 = 3.215$$

Local and Global Miss Rates, Working Set

local miss rate

It is equal to the number of misses in a cache at level i divided by the total number of accesses at level i.

global miss rate

It is equal to the number of misses in a cache at level i divided by the total number of memory accesses.

working set

The amount of memory, a given program [illegible]es in a time interval.

AMAT for Multi-Level Cache

- Assuming, a memory system with an L1 and L2 cache, then

$$\begin{aligned} AMAT &= L1_{hit\ time} + L1_{miss\ rate} \times L1_{miss\ penalty} \\ &= L1_{hit\ time} + L1_{miss\ rate} \times (L2_{hit\ time} + L2_{miss\ rate} \times L2_{miss\ penalty}) \end{aligned}$$

- All the memory accesses need to access the L1 cache irrespective of a hit or a miss. Hence, they need to incur a delay equal to $L1_{hit\ time}$.
- A fraction of accesses, $L1_{miss\ rate}$, will miss in the L1 cache, and move to the L2 cache. Irrespective of a hit or a miss, we need to incur a delay of $L2_{hit\ time}$ cycles.
- If a fraction of accesses ($L2_{miss\ rate}$) miss in the L2 cache, then they need to proceed to main memory.
- We have assumed that all the accesses hit in the main memory. Hence, the $L2_{miss\ penalty}$ is equal to the main memory access time.

n-Level Memory System

$$AMAT = L1_{hit\ time} + L1_{miss\ rate} * L1_{miss\ penalty}$$

$$L1_{miss\ penalty} = L2_{hit\ time} + L2_{miss\ rate} * L2_{miss\ penalty}$$

$$L2_{miss\ penalty} = L3_{hit\ time} + L3_{miss\ rate} * L3_{miss\ penalty}$$

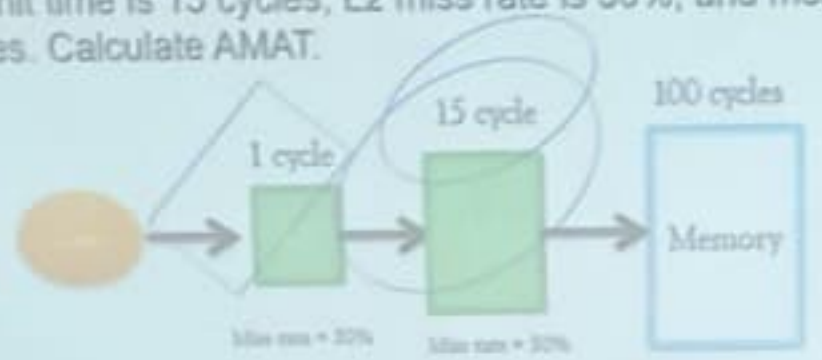
$$\ldots = \ldots$$

$$L(n-1)_{miss\ penalty} = Ln_{hit\ time}$$

- Factor Influencing:
  - Local Miss Rate
  - Global Miss Rate
  - Working Set

AMAT for Multi-Level Cache

- **Example:** Assume we have a two-level cache hierarchy system that runs an application. If you know that L1 cache hit time is 1 cycle, L1 miss rate is 30%, L2 hit time is 15 cycles, L2 miss rate is 50%, and memory latency is 100 cycles. Calculate AMAT.



- AMAT = Hit Time L1 + Miss Rate L1 X (Hit Time L2 + Miss Rate L2 X Miss penalty)
- AMAT = 1 + 30% X (15 + 50% X 100) = 20.5 cycles